QUALCOMM SNAPDRAGON® X ELITE TECHNICAL WHITEPAPER
Title: Private On-Device AI Architecture for Next-Generation Windows PCs
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1. EXECUTIVE SUMMARY & MOTIVATION
Modern desktop computing is undergoing a structural paradigm shift toward on-device neural processing. Traditional cloud-

The Snapdragon® X Elite processor revolutionizes personal computing by introducing a dedicated Qualcomm® Hexagon· N

2. CORE HARDWARE PILLARS
· Dedicated Hexagon NPU: 45 TOPS dedicated neural acceleration isolated from the CPU/GPU.
· Qualcomm Oryon· CPU: 12 high-performance cores delivering workstation-class responsiveness with ultra-low thermal dis
· Unified Memory Subsystem: Up to 64GB LPDDR5x memory operating at 8448 MT/s, eliminating PCI-e transfer bottleneck
· Power Efficiency: Up to 3x higher performance-per-watt compared to legacy x86 architectures, ensuring multi-day battery

3. SOFTWARE ARCHITECTURE: QUALCOMM AI HUB & QNN EXECUTION PROVIDER
Qualcomm AI Hub serves as the developer gateway for deploying state-of-the-art foundation models directly to Snapdragon

Key software components include:
1. Qualcomm AI Runtime (QAIRT): Low-level hardware drivers interfacing directly with the Hexagon Tensor Processor (HTP
2. ONNX Runtime QNN Execution Provider (QNN EP): Enables native execution of ONNX model graphs directly on the NPU
3. Qualcomm AI Hub Model Catalog: Pre-quantized and verified INT4/W4A16 models including Llama-3.2-1B/3B, Phi-3.5-m

4. PRIVACY AND LOCAL DATA SOVEREIGNTY
By confining embedding generation and token decoding to on-chip SRAM and unified LPDDR5x memory, sensitive docume

5. BENCHMARK RESULTS & BATTERY EFFICIENCY

Empirical testing on HP Snapdragon-powered laptops demonstrates dramatic improvements over CPU-only fallbacks:

· First-Token Latency: Reduced from 420ms (x86 CPU) down to 32ms on Hexagon NPU.

· Document Embedding Throughput: 1,450 chunks per second using INT8 quantized MiniLM.

· System Power Draw: Average 4.2 Watts under continuous generation vs 38 Watts on discrete GPU laptops.

· Offline Resilience: 100% functionality maintained with zero internet connectivity.

6. CONCLUSION & STRATEGIC RECOMMENDATIONS

On-device AI workspaces represent the definitive standard for confidential knowledge management. Developers must targe

7. ACTION ITEMS & SUBMISSION DEADLINES

For the Qualcomm Snapdragon AI Lab Build & Present Challenge 2026, the team must execute the following critical milesto

· Action Item 1: Complete end-to-end testing of local document parsing and RAG citations before October 10, 2026.
· Action Item 2: Validate QNN Execution Provider fallback mechanisms across both ARM64 Snapdragon and x86_64 host m
· Action Item 3: Deliver comprehensive Privacy Audit logs proving zero cloud API calls and zero network bytes transmitted.
· Action Item 4: Record final 3-minute video presentation and publish public GitHub repository by October 15, 2026.
· Action Item 5: Review benchmark metrics on target HP OmniBook device prior to final jury evaluation.